

Lecture (2): Muscles of the neck

Anterior triangle of the neck: It is bounded by:

- **Anterior border of the sternomastoid muscle**, posteriorly.
- **Midline of the neck**, anteriorly.
- **Lower border of the mandible**, superiorly.

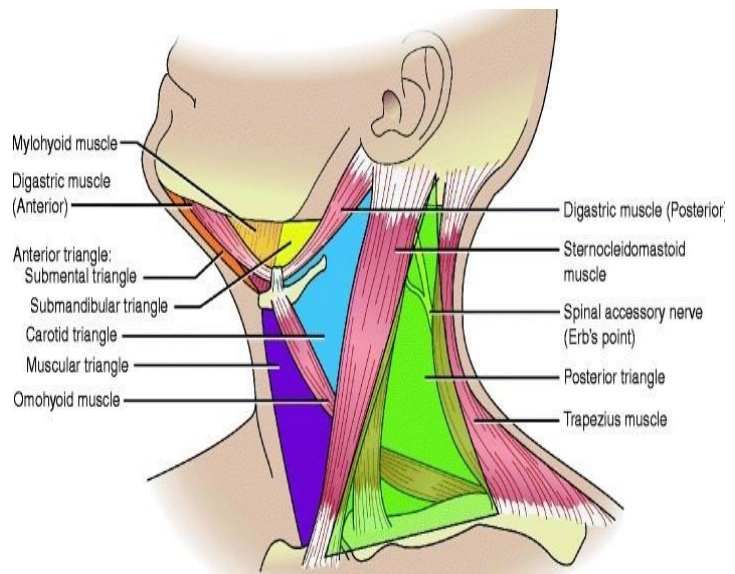
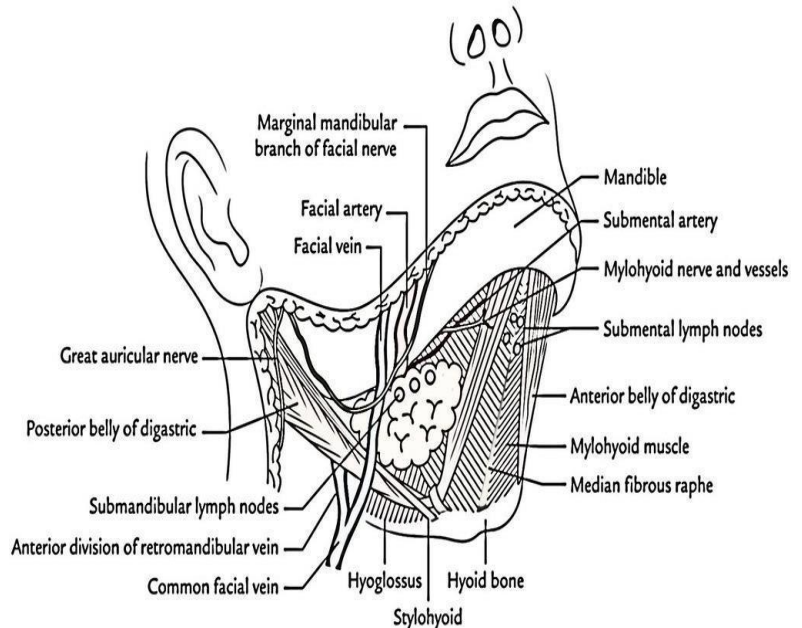
Subdivisions of the anterior triangle:

1- Digastric (submandibular):

- **Boundaries:** Inferior margin of the mandible, anterior belly of the digastric, posterior belly of the digastric muscle.
- **Contents:** Hypoglossal nerve (CN XII), nerve to mylohyoid, facial artery and anterior facial vein, submandibular gland lymph nodes.

2- Carotid triangle:

- **Boundaries:** Posterior belly of the digastric, superior belly of the omohyoid and anterior border of the sternocleidomastoid.
- **Contents:** common carotid, internal carotid and external carotid arteries. superior thyroid artery, lingual artery, facial artery and occipital pharyngeal arteries. Vagus nerve and its superior laryngeal branch, hypoglossal



and ansa cervicalis nerves, internal jugular vein and deep cervical lymph nodes.

3- Muscular triangle:

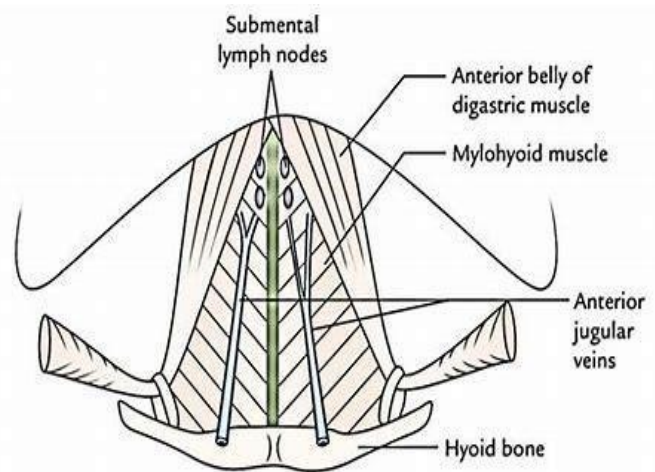
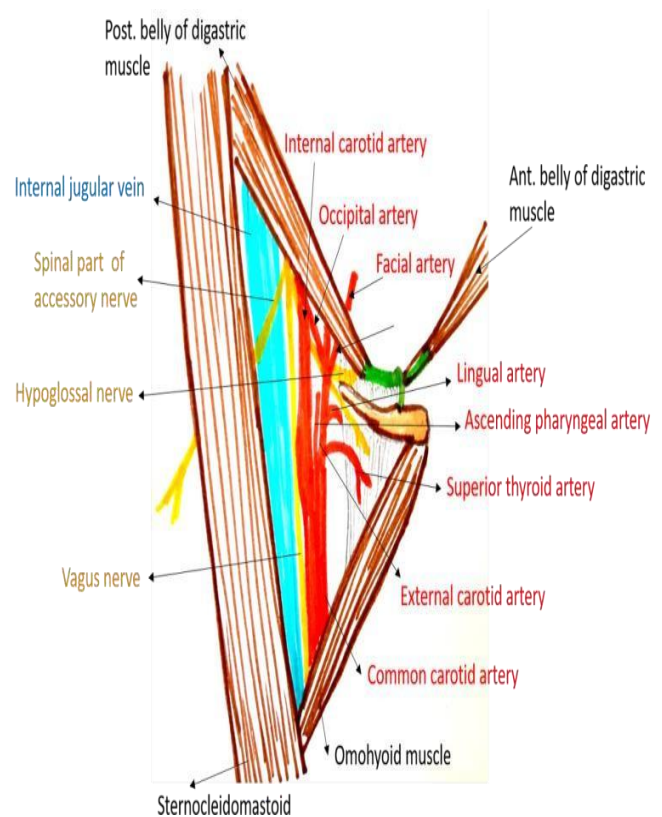
- **Boundaries:** Midline of the neck, superior belly of the omohyoid and the sternocleidomastoid muscle
- **Contents:** strap muscles of the neck (sternohyoid & sternothyroid muscles) that cover thyroid gland.

4- Submental triangle:

- **Boundaries:** body of **hyoid bone**, anterior belly of the digastric and the midline.
- **Contents:** Submental lymph nodes & submental veins.

Digastric muscle:

- **Origin:** the anterior belly arises from the digastric fossa at the lower border of mandible close to the midline. The posterior belly arises from the mastoid notch.
- **Insertion:** both are inserted into the intermediate tendon which suspends the hyoid bone by fibrous ring.
- **Nerve supply:** the anterior belly is supplied by the mandibular nerve & the posterior belly is supplied by the facial nerve.
- **Action:** The muscle elevates hyoid during swallowing.



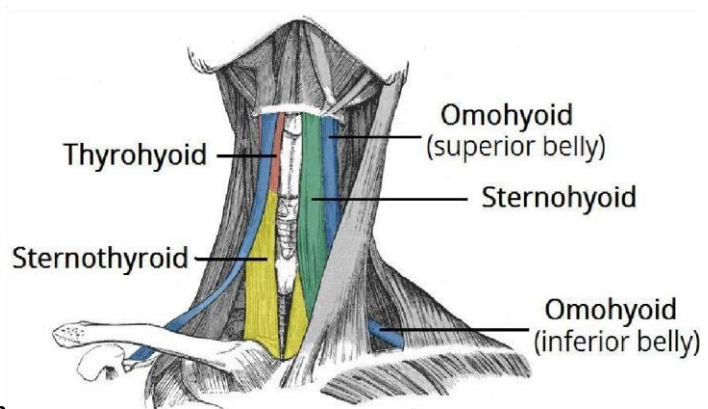
Mylohyoid muscle (diaphragma oris):

- It forms the floor of the oral cavity (oral diaphragm).
- **Origin:** Mylohyoid line at the inner side of the body of the mandible.
- **Insertion:** into Mylohyoid raphe, body of hyoid bone.
- **Nerve supply:** nerve to mylohyoid from mandibular nerve.
- **Action:** Elevates hyoid bone during swallowing and depresses the mandible when hyoid is fixed.

Infra-hyoid muscles:

- They are **strap muscles** of the neck lying below the hyoid bone.
- All infra-hyoid muscles are supplied by the **ansa cervicalis** except the **thyrohyoid muscle** which is supplied by the **C1** fibers via the **hypoglossal nerve**.
- During deglutition (swallowing), the strap muscles depress the hyoid bone and the thyroid cartilage.
- The **sternohyoid** arises from the posterior surface of manubrium & **inserted** into the lower border of hyoid close to the midline.

- The **sternothyroid muscle** arises from the posterior surface of the manubrium & **inserted** into the oblique line of the thyroid cartilage.

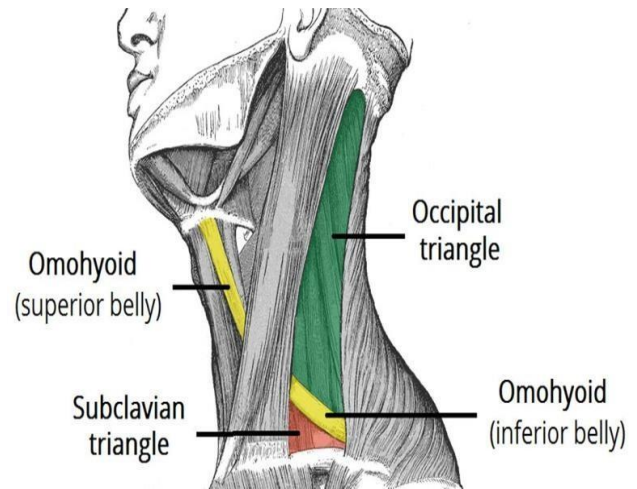


- **Thyrohyoid** arises from the thyroid cartilage & extends to the hyoid bone.
- **The omohyoid muscle** has an **inferior belly** that arises from the scapula medial to the suprascapular notch and insert into the intermediate (central) tendon. The **superior belly** arises from the central tendon and inserts into the hyoid bone. The central tendon is found deep in the sternocleidomastoid muscle.

Posterior triangle of the neck:

■ **Boundaries:**

- **Anterior:** Posterior margin of sternocleidomastoid muscle.
- **Posterior:** Anterior margin of trapezius muscle.
- **Inferior:** Middle one-third of clavicle.
- **Floor:** Splenius capitis, levator scapulae, Scalenus medius.
- **Roof:** skin, SF and deep investing fascia of the neck.

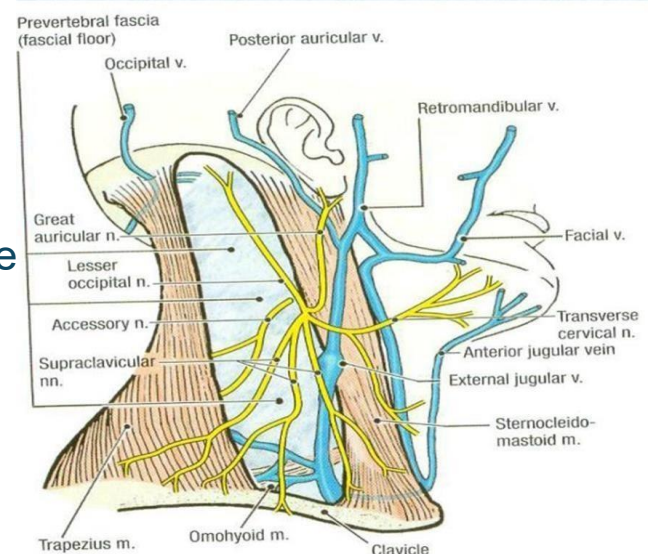


■ **Subdivisions:**

- The **inferior belly of the omohyoid muscle** divides the posterior triangle into the **occipital and subclavian triangles**.
- The **occipital (neural) triangle** is superior to the omohyoid Muscle.
- The **subclavian (vascular) triangle** is inferior to the omohyoid muscle.

■ **Contents:**

- **Nerves:** Trunks of the brachial plexus, Branches of the cervical plexus, Spinal root of the accessory nerve.
- **Arteries:** 3rd part of subclavian artery, Transverse cervical artery, Suprascapular artery & Occipital artery.
- **External jugular vein & subclavian vein.**
- **Supraclavicular and occipital lymph nodes.**
- **Inferior belly of omo-hyoid muscle.**



Sternocleidomastoid muscle:

- It divides the side of the neck into **anterior and posterior triangles**.

Origin:

1- **Clavicular head:** from upper surface of **medial 1/3 of clavicle**.

2- **Sternal head:** from the front of the **manubrium sterni**.

Insertion:

1- The lateral surface of the mastoid process.

2- Lateral 1/2 of the superior nuchal line.

Nerve supply:

1- The **spinal root of accessory nerve**.

2- **Cervical nerves C2, C3.**

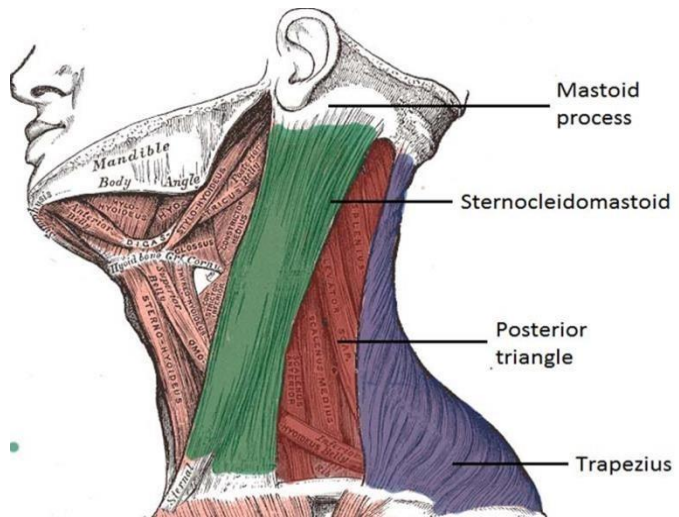
Action:

1- One muscle **bends the neck to its own side & rotates the face to the opposite side**.

2- Both sides: **Neck flexion**.

- **Its superficial relations** are skin, platysma, external jugular vein and parotid gland.

- **Its deep relations** are intermediate tendon of omohyoid, anterior jugular vein, carotid sheath and its contents, deep cervical lymph nodes, phrenic nerve and spinal accessory nerve.



Scalene muscles:

- The scalene muscles **flex the neck** to the same side. With both sides working together, the scalene muscles can **flex the neck forward** on the chest.

- The scalene muscles can also act as accessory respiratory (inspiratory) muscles by elevating the first and second ribs.

- The scalene muscles are innervated by the **lower cervical (C4–C8) spinal nerves**.

- The **anterior scalene muscle** arises from the **anterior tubercles of the transverse processes of C3–C6**. It inserts into the **scalene tubercle of the first rib**.

- The **middle scalene muscle** is the largest of the scalene muscles and arises from

